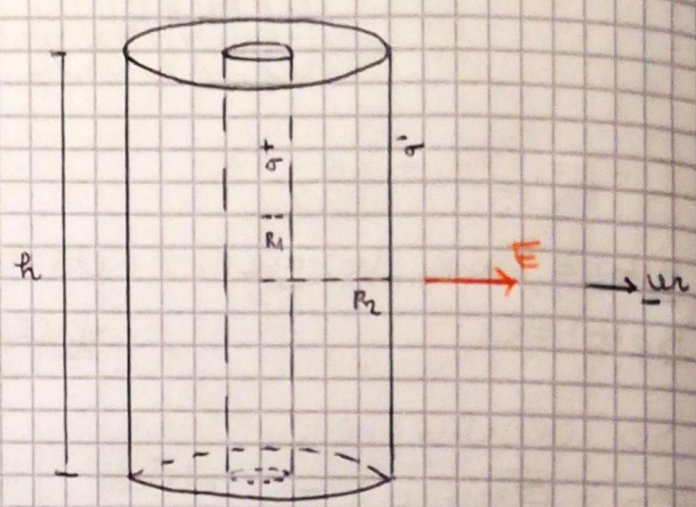
APPELLO 20/06/2018

ESERCIZIO 1

Considero un tratto di altezza h. Allora:

$Q^+ = \sigma 2\pi R_1 h$

$Q^- = -\sigma 2\pi R_2 h$



SE $0 < r < R_1$

$\underline{E} = 0$

SE $R_1 < r < R_2$

$$\cancel{2\pi} r \cancel{h} E = \frac{\sigma \cancel{2\pi} R_1 \cancel{h}}{\varepsilon_0}$$

$$\Rightarrow \underline{E} = \frac{\sigma R_1}{\varepsilon_0 r} \underline{u}_r$$

SE $r > R_2$

$$Q_{INT} = Q^+ + Q^- = 2\pi\sigma h (R_1 - R_2) < 0$$

$$\cancel{2\pi} r \cancel{h} E = \frac{\cancel{2\pi}\sigma \cancel{h}}{\varepsilon_0}(R_1 - R_2)$$

$$\underline{E} = \frac{\sigma(R_1 - R_2)}{\varepsilon_0 r} \underline{u}_r$$

$$\Delta V = V(R_2) - V(R_1) = -\int_{R_1}^{R_2} \frac{\sigma R_1}{\varepsilon_0 r} dr = \frac{\sigma R_1}{\varepsilon_0} \ln \frac{R_1}{R_2}$$

$$\cancel{U = \int_\tau u_m d\tau = \int_{R_1}^{R_2} \frac{\sigma R_1}{\varepsilon_0 r} 2\pi r h \, dr =}$$

$$\cancel{= \frac{\sigma R_1 2\pi h}{\varepsilon_0}(R_2 - R_1)}$$

$$U = \int_\tau u_m d\tau$$

$$u_m = \frac{1}{2}\varepsilon_0 \frac{\sigma^2 R_1^2}{\varepsilon_0^2 r^2} = \frac{\sigma^2 R_1^2}{2\varepsilon_0 r^2}$$

$$U = \int_{R_1}^{R_2} \frac{\sigma^2 R_1^2 \pi h}{\varepsilon_0 r} dr = \frac{\pi \sigma^2 R_1^2 h}{\varepsilon_0} \ln \frac{R_2}{R_1}$$

$$\Rightarrow \frac{U}{h} = \frac{\pi \sigma^2 R_1^2}{\varepsilon_0} \ln \frac{R_2}{R_1}$$

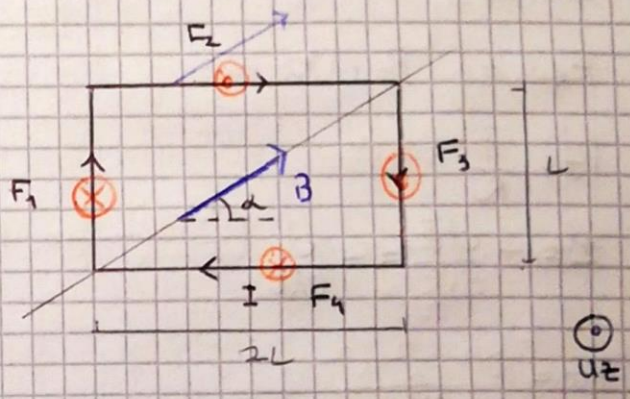
ESERCIZIO 2

$\underline{F} = I \oint_\gamma \underline{u}_t \times \underline{B}\, dl$

c)

Supponendo che una corrente I scorra in verso orario

$\text{sen}\,\alpha = \frac{1}{\sqrt{5}}$



$\underline{F}_1 = -BIL\cos\alpha\, \underline{u}_z$

$\underline{F}_2 = BI2L\,\text{sen}\,\alpha\, \underline{u}_z$

$\underline{F}_3 = BIL\cos\alpha\, \underline{u}_z$

$\underline{F}_4 = -BI2L\sin(\pi-\alpha)\, \underline{u}_z = -BI2L\,\text{sen}\,\alpha\, \underline{u}_z$

la forza risultante totale è nulla.

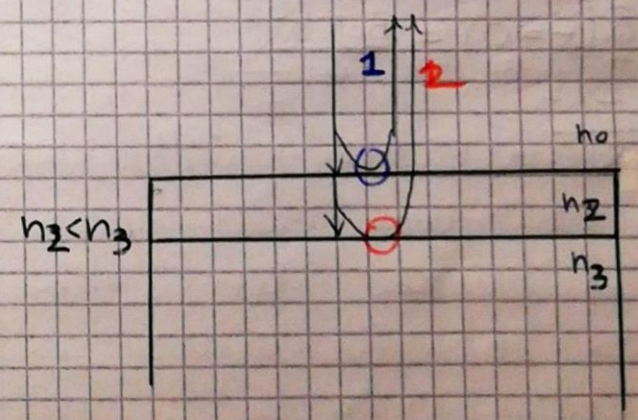
ESERCIZIO 3

$\cancel{\Delta\varphi = J \neq f \frac{\partial\varphi}{\partial t}}$

$\Delta\varphi = \varphi_2 - \varphi_1$

$\varphi_1 = \pi$ perché $n_0 < n_2$

$\varphi_2 = \pi + 2dk = \pi + 2d\frac{2\pi n_2}{\lambda}$

$\Rightarrow \Delta\varphi = \frac{2d\pi n_2}{\lambda_0} = (2m+1)\pi$

$d = \frac{(2m+1)}{4}\frac{\lambda_0}{n_2}$